

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

L-2/T-II CSE 220 Signals and Linear Systems Sessional

Time: 40 mins

Marks: 30

Student Name: Kazi Asef KabiraStudent ID: 2305084

There are 6 questions in total. Answer all the questions in the spaces provided on the question sheets. No extra sheets will be provided

1. What is the output of the following Python program?

(5)

```
import numpy as np
arr = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9])
arr = arr[arr >= 5] → values ≥ 5
print(arr)
arr = arr[arr % 2 != 0] → odd values
print(arr)
arr = arr ** 2 → square
print(arr)
arr = arr % 10 → last digit
print(arr)
result = arr.mean() → mean
print(result)
```

Write down the Output below:

[5, 6, 7, 8, 9]

[5, 7, 9]

[25, 49, 81]

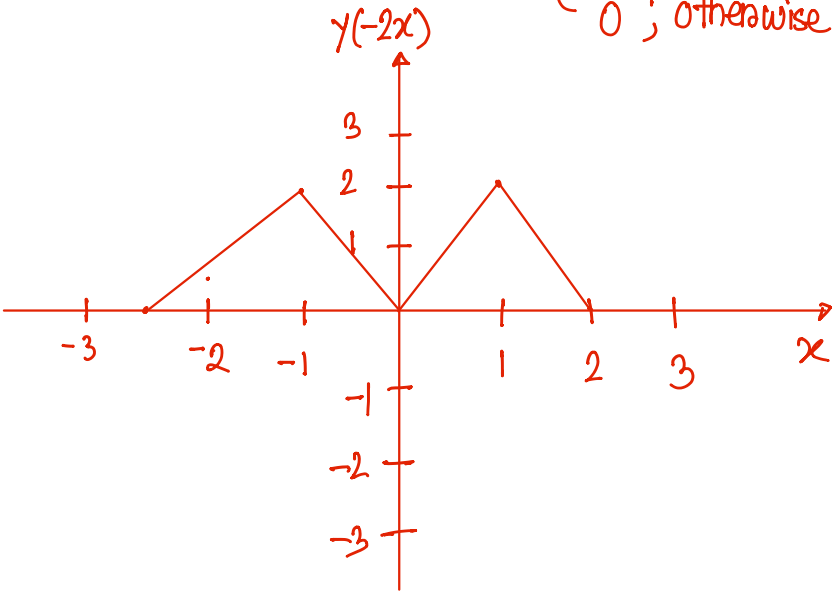
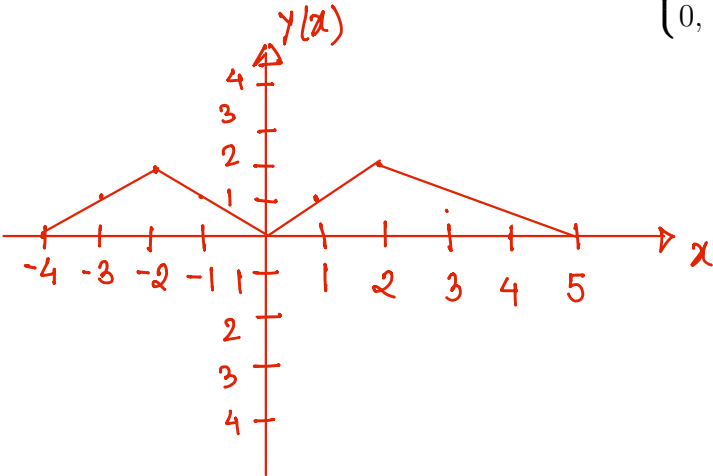
[5, 9, 1]

5.0

2. Plot the signals $y(x)$ and $y(-2x)$.

$$y(x) = \begin{cases} 4 + x, & -4 < x < -2 \\ -x, & -2 < x < 0 \\ x, & 0 < x < 2 \\ \frac{10}{3} - \frac{2}{3}x, & 2 < x < 5 \\ 0, & \text{otherwise} \end{cases}$$

$$y(-2x) = \begin{cases} 4 - 2x; & -1 < x < 2 \\ 2x; & 0 < x < 1 \\ -2x; & -1 < x < 0 \\ \frac{10}{3} + \frac{4}{3}x; & -\frac{5}{2} < x < -1 \\ 0; & \text{otherwise} \end{cases}$$



3. An input $x[n]$ is applied to two LTI systems in parallel: (5)

$$h_1[n] = 2\delta[n] - \delta[n-1] + \delta[n-3], \quad h_2[n] = \delta[n] + \delta[n-2] - 2\delta[n-4]$$

The outputs of these two branches are added, and the resulting signal is then passed through a third LTI system in series with impulse response

$$h_3[n] = \delta[n] - 2\delta[n-1] + \delta[n-2].$$

- (a) Find the equivalent impulse response of the overall system.
 (b) For $x[n] = u[n]$, compute the output of the overall system in terms of $u[n]$.

$$\textcircled{a} \quad h_p = h_1 + h_2 = 3\delta[n] - \delta[n-1] + \delta[n-2] + \delta[n-3] - 2\delta[n-4]$$

$$h_p = [3, -1, 1, 1, -2]$$

$$h_3 = [1, -2, 1]$$

$$h_{eq} = h_p * h_3$$

$$= [3, -7, 6, -2, -3, 5, -2]$$

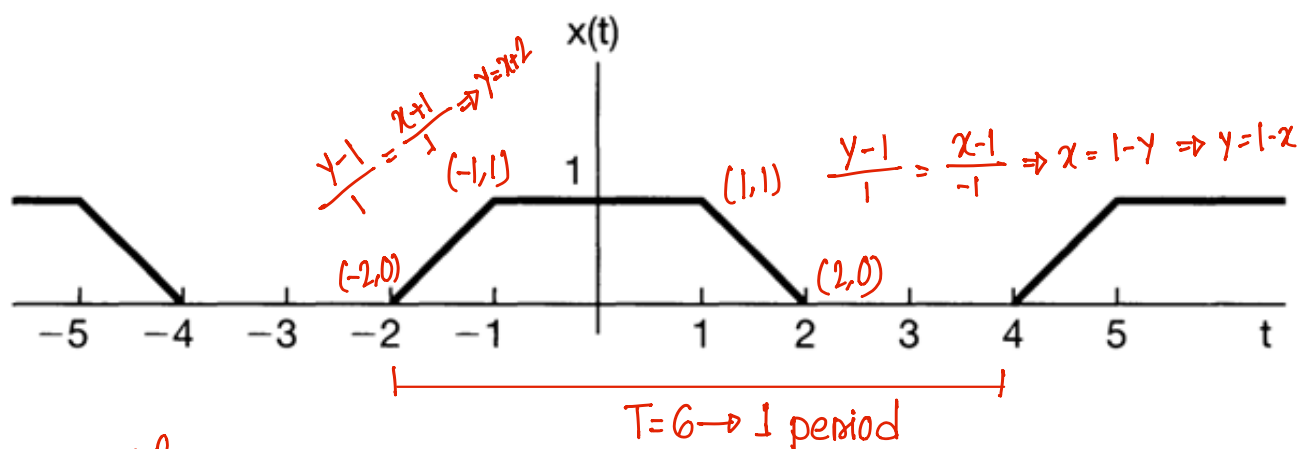
$$\textcircled{b} \quad y[n] = x[n] * h_{eq}[n] = 3\delta[n] - 7\delta[n-1] + 6\delta[n-2] - 2\delta[n-3]$$

$$= 3u[n] - 7u[n-1] + 6u[n-2] - 2u[n-3]$$

$$- 3u[n-4] + 5u[n-5] - 2u[n-6]$$

4. Determine the Fourier series representation for the following signal:

(5)



$$a_0 = \frac{1}{6} \int_{-2}^2 x(t) dt$$

$$= \frac{1}{6} \left[\int_{-2}^{-1} (t+2) dt + \int_{-1}^1 1 dt + \int_1^2 (2-t) dt \right] = 1/2$$

$$x(t) = \begin{cases} t+2; & -2 \leq t \leq -1 \\ 1; & -1 \leq t \leq 1 \\ 2-t; & 1 \leq t \leq 2 \\ 0; & 2 \leq t \leq 4 \end{cases}$$

$$a_k = \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt = \frac{1}{6} \left[\int_{-2}^{-1} (t+2) e^{-jk\omega_0 t} dt + \int_{-1}^1 e^{-jk\omega_0 t} dt + \int_1^2 (2-t) e^{-jk\omega_0 t} dt \right]$$

$$= \frac{1}{6} \left[\int_{-2}^{-1} t e^{-jk\omega_0 t} dt + \int_{-2}^{-1} 2 e^{-jk\omega_0 t} dt + \int_{-1}^1 e^{-jk\omega_0 t} dt + 2 \int_1^2 e^{-jk\omega_0 t} dt - \int_1^2 t e^{-jk\omega_0 t} dt \right]$$

$$= \frac{1}{6} \left[\frac{e^{-jk\omega_0 t}}{-jk\omega_0} \right]_{-2}^{-1} + \frac{1}{3} \left[\frac{e^{-jk\omega_0 t}}{-jk\omega_0} \right]_{-2}^{-1} + \frac{1}{6} \left[\frac{e^{-jk\omega_0 t}}{-jk\omega_0} (t+1) \right]_{-1}^1$$

$$- \frac{1}{6} \left[\frac{-e^{-jk\omega_0 t}}{-jk\omega_0} (t+1) \right]_1^2$$

$$= \frac{1}{-6jk\omega_0} (e^{-j2k\omega_0} - e^{j2k\omega_0}) - \frac{1}{3jk\omega_0} (e^{jk\omega_0} - e^{j2k\omega_0}) + \frac{1}{6jk\omega_0} (2e^{-jk\omega_0} + e^{j2k\omega_0})$$

$$- \frac{1}{6jk\omega_0} (3e^{-j2k\omega_0} - 2e^{-jk\omega_0})$$

$$= \frac{6}{n^2 \pi^2} \left(\cos \frac{n\pi}{3} - \cos \frac{2n\pi}{3} \right)$$

5. A rectangular signal is defined as

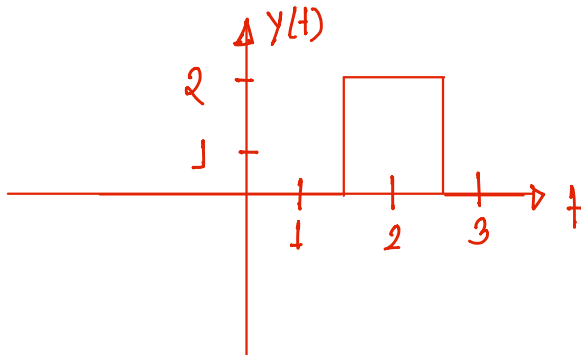
(5)

$$x(t) = \begin{cases} 1, & |t| \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Consider the signal $y(t) = 2x^*(2t - 4)$ where $*$ denotes the **complex conjugate** operation.

(a) Sketch the signal $y(t)$.

$$\begin{aligned} x(t) \text{ is real} &\Rightarrow x^*(t) = x(t) & \text{None zero} &\Rightarrow |2t - 4| \leq 1 \\ & & &\Rightarrow -1 \leq 2t - 4 \leq 1 \\ & & &\Rightarrow \frac{3}{2} \leq t \leq \frac{5}{2} \end{aligned}$$



(b) Determine the Continuous Fourier Transform $Y(\omega)$.

$$\begin{aligned} X(\omega) &= \int_{-1}^1 e^{-j\omega t} dt = \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-1}^1 = \frac{e^{-j\omega} - e^{j\omega}}{-j\omega} \\ &= \frac{2\sin\omega}{\omega} \end{aligned}$$

$$x(t) \longleftrightarrow X(\omega) = \frac{2\sin\omega}{\omega}$$

$$x^*(t) \longleftrightarrow X^*(\omega) = \frac{2\sin\omega}{\omega}$$

$$x\left(\frac{t}{2}\right) \longleftrightarrow \frac{1}{2} X\left(\frac{\omega}{2}\right) = \frac{\sin\omega/2}{\omega/2} = \frac{2\sin\omega/2}{\omega}$$

$$x(2t+2) \longleftrightarrow e^{-j2\omega} \cdot \frac{2\sin\omega/2}{\omega}$$

$$2x(2t+2) \longleftrightarrow \frac{4\sin\omega/2}{\omega} e^{-j(2\omega)}$$

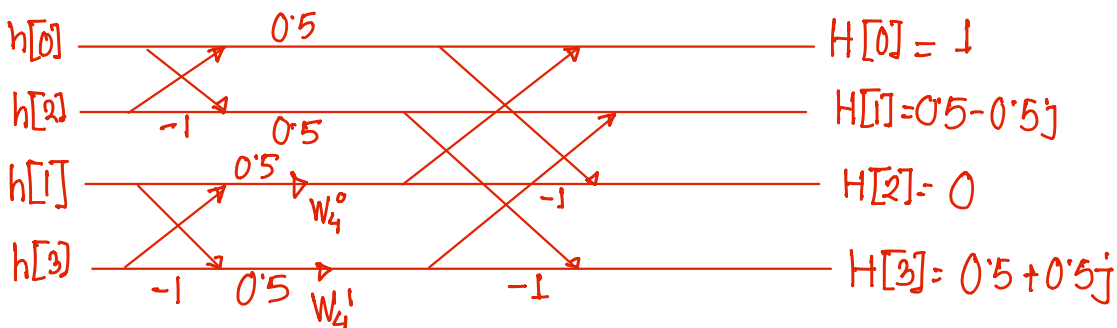
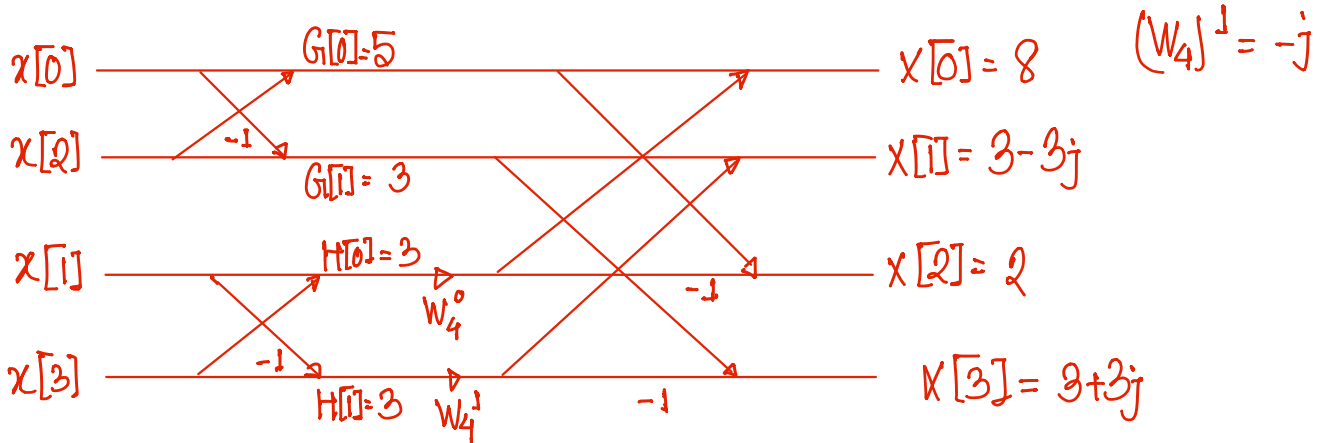
$$Y(\omega) = \frac{4\sin\omega/2}{\omega} e^{-j2\omega}$$

6. You are given a signal $x[n] = [4, 3, 1]$. You want to apply a filter defined by the impulse response $h[n] = [0.5, 0.5]$. (5)

(a) Using a 4-point Fast Fourier Transform (FFT), find the frequency spectrum of the output signal.

zero pad = $3+2-1 = 4 \Rightarrow x[n] = [4, 3, 1, 0]$, $h[n] = [0.5, 0.5, 0, 0]$

4 point twiddle $\Rightarrow W_4 = e^{-j\frac{2\pi}{4}} = -j$ = $[4, 3, 1, 0]$ $(W_4)^0 = 1$



$Y[k] = X[k] H[k] = [8, -3j, 0, 3j]$

(b) Compare this to the frequency spectrum of the original signal $x[n]$. Based on your comparison, comment on the nature of the filter $h[n]$ (e.g., low-pass, high-pass).

k	0	1	2 ^(fs/2)	3
$ X[k] $	8	$3\sqrt{2}$	2	$3\sqrt{2}$
$ H[k] $	1	$1/\sqrt{2}$	0	$1/\sqrt{2}$
$ Y[k] $	8	3	0	3

High frequency is completely removed $\rightarrow$ low pass

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There are 8 questions in total. Answer all the questions in the spaces provided on the question sheets. No extra sheets will be provided

1. What is the output of the following Python program?

(5)

```
import numpy as np
```

```
data = np.array([3, 6, 1, 2, 4, 7, 8])
```

```
data = data[ data % 2 == 0 ]
```

```
data = data / 2
```

```
result = data ** 2
```

```
print(result.sum())
```

Write down the Output below:

30.0

2. What is the output of the following Python program?

(5)

```
projects = {
    "Project_A": {"priority": 1, "team_performance": 85, "deadline": 15},
    "Project_B": {"priority": 2, "team_performance": 70, "deadline": 20},
    "Project_C": {"priority": 3, "team_performance": 90, "deadline": 10},
    "Project_D": {"priority": 1, "team_performance": 60, "deadline": 25},
    "Project_E": {"priority": 3, "team_performance": 75, "deadline": 18},
    "Project_F": {"priority": 3, "team_performance": 80, "deadline": 12},
    "Project_G": {"priority": 1, "team_performance": 88, "deadline": 8},
    "Project_H": {"priority": 1, "team_performance": 65, "deadline": 30},
}

selected_projects = list(projects.items())[1::3]
total_resources = 0

for i, (project, details) in enumerate(selected_projects):
    allocated_resources = 0

    if details["priority"] == 1:
        allocated_resources += 50
    elif details["priority"] == 2:
        allocated_resources += 30
    else:
        allocated_resources += 20

    if details["team_performance"] > 80:
        allocated_resources += 20
    elif details["team_performance"] > 60:
        allocated_resources += 10

    if i % 2 == 0:
        allocated_resources += 10

    if details["deadline"] <= 15:
        allocated_resources += 15
    else:
        allocated_resources += 5

    total_resources += allocated_resources
    print(f"Resources allocated for {project}: {allocated_resources}")

total_resources += 50

print("Total allocated resources:", total_resources)
```

Write down the Output below:

Resources allocated for Project_B: 55
Resources allocated for Project_C: 65
Resources allocated for Project_D: 75
Total allocated resources: 245

3. A temperature sensor in a remote location collects data samples of the temperature at a rate of **3 samples per second**, forming a discrete-time signal $x[n]$, where n represents the sample index. The data needs to be transmitted to a central monitoring system via a low-bandwidth satellite link that can only handle a maximum rate of **2 samples per second**. As a result, the exact signal cannot be transmitted to the central server. But the central system requires the data at the original rate of **3 samples per second** for accurate analysis and visualization. (5)

Question I: Explain a way to send the signal through the satellite link and reconstruct the signal at the central server. The reconstructed signal will be an approximation of the original signal.

Question II: For the discrete-time signal over the first 3 seconds:

$$x[n] = \{0.0, 0.5, 0.87, 1.0, 0.87, 0.5\}, \quad n = 0, 1, 2, 3, 4, 5$$

1. What is the **input signal to the channel**?
2. What is the **reconstructed signal** at the receiver ?

4. Suppose you have three signals $a[n]$, $b[n]$ and $c[n]$ defined below. We want to generate three signals $x[n]$, $y[n]$ and $z[n]$. Now determine if it is possible to express each of $x[n]$, $y[n]$ and $z[n]$ as the result of convolution between any two signals among $a[n]$, $b[n]$ and $c[n]$ (e.g. $a * b$). If it is, then express them as a result of convolution. Otherwise, explicitly mention that it is not possible. Assume that the signal values are padded with zeroes. (5)

$$x = a * b$$

$$y = a * c$$

$$z = b * b$$

$$a[n] = [1, 0, 1, 1] = 1 + z^2 + z^3 \quad \sum a = 3$$

$$b[n] = [1, 1, 0, 1] = 1 + z + z^3 \quad \sum b = 3$$

$$c[n] = [1, 1, 1, 1] = 1 + z + z^2 + z^3 \quad \sum c = 4$$

$$x[n] = [1, 1, 1, 3, 1, 1, 1] \quad \sum x = 9$$

$$y[n] = [1, 1, 2, 3, 2, 2, 1] \quad \sum y = 12$$

$$z[n] = [1, 2, 1, 2, 2, 0, 1] \quad \sum z = 9$$

5. Two windmills, **Windmill A** and **Windmill B**, generate power based on the wind's direction (x): (5)

- **Windmill A:** Generates power that depends only on the **magnitude of the wind speed** and is unaffected by changes in the wind's direction.
- **Windmill B:** Generates power that depends on both the **speed and direction of the wind** and flips sign when the wind direction reverses.

The total power generated by the two windmills is modeled as:

$$P(x) = x^3 + 3 \cos(x) - 5x^2 + \sin(2x).$$

$$P(-x) = -x^3 + 3 \cos x - 5x^2 - \sin 2x$$

Question 1: Separate and Verify Power Contributions

1. **Separate the Contributions:** Decompose $P(x)$ into its **even** and **odd** components:

- The **even component** corresponds to the contribution from **Windmill A** (unaffected by wind direction reversal).
- The **odd component** corresponds to the contribution from **Windmill B** (flips sign with wind direction reversal).

2. **Verify:** Show that the combined power $P(x)$ can be reconstructed by summing the even and odd components.

$$1. P_e(x) = \frac{1}{2} [P(x) + P(-x)]$$

$$= 3 \cos x - 5x^2$$

$$P_o(x) = \frac{1}{2} [P(x) - P(-x)]$$

$$= x^3 + \sin 2x$$

$$2. P_e(x) + P_o(x) = x^3 + 3 \cos x - 5x^2 + \sin 2x$$

Question 2: Fourier Series Analysis

1. **For Windmill A (Even Component):** Which Fourier series coefficients (A_n or B_n) will be zero for its power contribution? Explain.
2. **For Windmill B (Odd Component):** Which Fourier series coefficients (A_n or B_n) will be zero for its power contribution? Explain.

$$1. B_n = \frac{2}{T} \int_{-T/2}^{T/2} P_o \sin(n \omega_0 x) dx = 0$$

$$2. A_0 = \frac{1}{T} \int P_o = 0 \quad A_n = \frac{2}{T} \int P_o \cos(n \omega_0 x) dx = 0$$

6. Find the fourier transform of e^{-t^2} . After that find the energy in time domain and frequency domain and show that they are equal. (5)

$$X(\omega) = \int_{-\infty}^{\infty} e^{-t^2} e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{-(t+j\omega/2)^2} e^{-\omega^2/4} dt = e^{-\omega^2/4} \int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi} e^{-\omega^2/4}$$

Time energy,
 $E = \int_{-\infty}^{\infty} e^{-2t^2} dt = \sqrt{\pi}/2$

frequency energy, $E = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X|^2 d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \pi e^{-\omega^2/2} d\omega = \sqrt{\pi}/2$

7. (a) The result of cross-correlation between 2 signals is given as $r[n] = [21.7, -29.2, 23.5, -27.6]$. You do not know the two original signals but you know that they are out of phase with respect to each other. Given all these information, find out the sample lag. (2)

$$r = [21.7, -29.2, 23.5, -27.6]$$

minimum is -29.2 at index=1

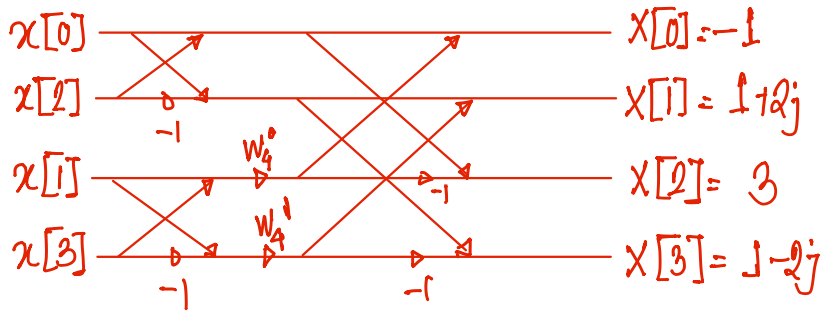
$\therefore$ Sample lag=1

$$W_4^0 = 1, \quad W_4^1 = -j$$

- (b) Suppose, two discrete signals $x[n] = [1, -2]$ and $y[n] = [2.5, -2.7, 3.4]$ are given. (3)
Calculate the frequency spectrum of the circular convolution of $x[n]$ & $y[n]$ using the 4-point Fast Fourier Transform (FFT).

$$x[n] = [1, -2, 0, 0]$$

$$y[n] = [2.5, -2.7, 3.4, 0]$$

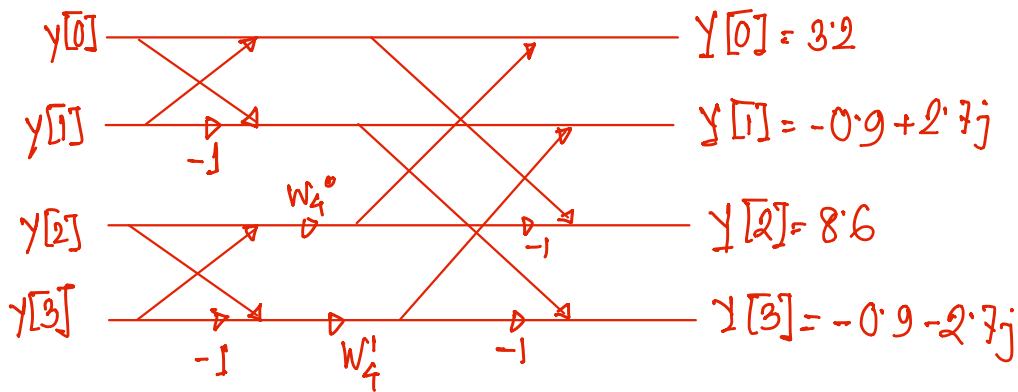


$$Z[0] = -3.2$$

$$Z[1] = -6.3 + 0.9j$$

$$Z[2] = 25.8$$

$$Z[3] = -6.3 - 0.9j$$



8. (a) You are given implementations of a CFT and an DFT algorithm. Is it possible to perform Laplace and z-transform using the given implementations without changing their code? If yes, describe how. If no, explain why. (5)

(b) You are given the following function defined on the set of integers:

$$x[n] = \begin{cases} 1, & \text{if } n \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Then formulate the function $x[-n-1]$ and draw the plot.

$$x[-n-1] = \begin{cases} 1 & ; n \leq -1 \\ 0 & ; n \geq 0 \end{cases}$$

9. (Bonus) Please write down some of the biggest challenges you faced in this course. Kindly suggest some ways that you think will be helpful for improving this course. (3)

Too much hard work

Sessional Quiz (Practice Set A)
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There are 8 questions in total. Answer all the questions in the spaces provided on the question sheets. No extra sheets will be provided.

1. What is the output of the following Python program? (5)

```
import numpy as np
x = np.array([2, 7, 4, 9, 6, 1, 8, 3]) → mean = 5
y = x[x > x.mean()]
print(y)
z = y.reshape(2, 2).T
print(z)
print(z.sum(axis=0))
w = np.cumsum(y) % 7
print(w)
print(w[w > 0].mean())
```

Write down the Output below:

[7, 9, 6, 8]		[6, 14]
[7, 6],		[0, 2, 1, 2]
[9, 8]		1.67

2. What is the output of the following Python program? (5)

```
students = {
    "Rafi": {"lab": 18, "quiz": 7, "attendance": 92}, ✓
    "Tanha": {"lab": 12, "quiz": 9, "attendance": 70},
    "Nabil": {"lab": 20, "quiz": 4, "attendance": 85}, ✓
    "Sadiah": {"lab": 15, "quiz": 8, "attendance": 60},
    "Arif": {"lab": 9, "quiz": 10, "attendance": 95}, ✓
    "Mim": {"lab": 17, "quiz": 6, "attendance": 78},
}
chosen = list(students.items())[:2]
bonus_pool = 0
for i, (name, s) in enumerate(chosen):
    score = s["lab"] + 2 * s["quiz"]
    if s["attendance"] >= 90:
        score += 5
    elif s["attendance"] >= 75:
        score += 2
    else:
        score -= 3
    if i == len(chosen) - 1:
        score *= 2
    if score > 35:
        bonus_pool += score - 35
print(f"{name}: {score}")
```

```
print("Bonus_pool:", bonus_pool)
```

Write down the Output below:

Rafi: 37

Nabil: 30

Arif: 68

Bonus_pool: 35

3. A continuous-time signal is given by

$$x(-t) = \begin{cases} -t; & 0 < t < 2 \\ 2; & -2 < t < -3 \\ 0; & \text{otherwise} \end{cases}$$

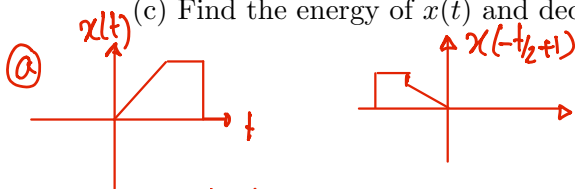
$$x(t) = \begin{cases} t, & 0 < t < 2 \\ 2, & 2 \leq t < 3 \\ 0, & \text{otherwise} \end{cases}$$

$$x(-t/2+1) = \begin{cases} 1-t/2; & -2 < t < 2 \\ 2; & -4 < t < -2 \\ 0; & \text{otherwise} \end{cases} \quad (5)$$

(a) Plot $x(t)$ and $x(-t/2+1)$.

(b) Find and sketch the even and odd components $x_e(t)$ and $x_o(t)$, and verify that $x_e(t) + x_o(t) = x(t)$ for $0 < t < 3$.

(c) Find the energy of $x(t)$ and deduce the energy of $x(-t/2+1)$ without further integration.



② $x_e(t) = \frac{1}{2} (x(t) + x(-t))$

$$= \begin{cases} \frac{|t|}{2}; & 0 < |t| < 2 \\ 1; & 2 \leq |t| < 3 \\ 0; & \text{otherwise} \end{cases}$$

$x_o(t) = \begin{cases} t/2; & |t| < 2 \\ 1; & 2 \leq |t| < 3 \\ -1; & -3 < t < -2 \\ 0; & \text{else} \end{cases}$

③ $E = \int_0^2 t^2 dt + \int_2^3 4 dt = 20/3$

$x(at+b) \Rightarrow E = \frac{E}{|a|}$

$= \frac{E}{1/2}$

$= 40/3$

4. Two LTI systems $h_1[n] = \delta[n] + 2\delta[n-1]$ and $h_2[n] = \delta[n] - \delta[n-1]$ are connected in series. This series combination is connected in parallel with a third LTI system $h_3[n] = \delta[n-2]$ (the two branch outputs are added). (5)

(a) Find the equivalent impulse response $h[n]$ of the overall system.

(b) Is the overall system causal? Is it stable? Is it memoryless? Justify each.

(c) Find the output for $x[n] = u[n] - u[n-3]$, and also for $x[n] = u[n]$ in terms of shifted $u[n]$.

5. A periodic sawtooth signal has period $T = 2$ and is given over one period by $x(t) = t$ for $-1 < t < 1$. (5)

(a) Determine its exponential Fourier series coefficients a_k .

CSE 220 Signals & Linear Systems

Complete Formula Sheet with Definitions

Signal properties · Convolution · Fourier series · CTFT · DTFT/DFT · FFT · Sampling · NumPy

Contents

1	Notation and basic signals	1
2	Signal operations and properties	2
3	System properties	3
4	Convolution and LTI systems	3
5	Continuous-time Fourier series	4
6	Continuous-time Fourier transform (CTFT)	5
7	Discrete-time Fourier transform (DTFT)	6
8	Discrete Fourier transform (DFT)	6
9	Fast Fourier transform (FFT)	8
10	Sampling and reconstruction	8
11	NumPy and Matplotlib reference	10
12	Useful mathematics	10

1 Notation and basic signals

Name	Formula	Definition / meaning
Continuous / discrete time	$x(t), t \in \mathbb{R}; \quad x[n], n \in \mathbb{Z}$	t is real time (s); n is the integer sample index. Round brackets = continuous, square brackets = discrete.
Euler's formula	$e^{j\theta} = \cos \theta + j \sin \theta$	Links complex exponentials to sinusoids; $j = \sqrt{-1}$. Gives $\cos \theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$, $\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$.
Unit step	$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases} \quad u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$	Switches a signal on at $t = 0 / n = 0$.
Unit impulse (CT)	$\delta(t) = 0 \ (t \neq 0), \int_{-\infty}^{\infty} \delta(t) dt = 1$	Infinitely narrow pulse of unit area; $\delta(t) = \frac{du}{dt}$.
Unit impulse (DT)	$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} = u[n] - u[n-1]$	Single sample of height 1; $u[n] = \sum_{k=-\infty}^n \delta[k]$.
Sifting property	$\int x(t)\delta(t-t_0)dt = x(t_0), \quad \sum_n x[n]\delta[n-n_0] = x[n_0]$	An impulse “picks out” the value of the signal at its location.
Impulse scaling	$\delta(at) = \frac{1}{ a } \delta(t)$	Compressing an impulse reduces its area.
Ramp	$r(t) = t u(t)$	Integral of the step; $\frac{dr}{dt} = u(t)$.
Rectangular pulse	$\text{rect}(t/\tau) = 1 \text{ for } t < \tau/2, \text{ else } 0$	Pulse of width τ , height 1, centred at 0.

Name	Formula	Definition / meaning
Triangular pulse	$\Lambda(t/\tau) = 1 - t /\tau$ for $ t \leq \tau$	Peak 1 at $t = 0$, base 2τ ; equals $\text{rect} * \text{rect}$ (scaled).
Sinc	$\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$, $\text{sinc}(0) = 1$	Normalised sinc (NumPy's <code>np.sinc</code>); zeros at nonzero integers.
Complex exponential	$x(t) = Ce^{st}$, $s = \sigma + j\omega$	σ sets growth/decay, ω sets oscillation (rad/s).
Sinusoid	$x(t) = A \cos(\omega_0 t + \phi)$, $\omega_0 = 2\pi f_0$	A amplitude, ω_0 angular frequency (rad/s), f_0 frequency (Hz), ϕ phase (rad).

2 Signal operations and properties

Name	Formula	Definition / meaning
Time shift	$y(t) = x(t - t_0)$	Delay (move right) if $t_0 > 0$; advance (left) if $t_0 < 0$.
Time reversal	$y(t) = x(-t)$	Mirror image about $t = 0$.
Time scaling	$y(t) = x(at)$	$ a > 1$ compresses, $ a < 1$ stretches, $a < 0$ also flips. Amplitude unchanged.
General transformation	$y(t) = x(at + b)$; breakpoint $u \mapsto t = \frac{u - b}{a}$	Each corner u of x moves to $t = (u - b)/a$ keeping its value. Equivalent: shift by b first, then scale by a .
DT decimation	$y[n] = x[Mn]$	Keeps every M -th sample; others are lost.
DT expansion	$y[n] = x[n/L]$ if $L \mid n$, else 0	Inserts $L - 1$ zeros between samples.
Even part	$x_e(t) = \frac{1}{2}[x(t) + x(-t)]$	Symmetric component: $x_e(-t) = x_e(t)$.
Odd part	$x_o(t) = \frac{1}{2}[x(t) - x(-t)]$	Antisymmetric: $x_o(-t) = -x_o(t)$, $x_o(0) = 0$. Always $x = x_e + x_o$.
Even/odd products	even $\times$ even = even, odd $\times$ odd = even, even $\times$ odd = odd	$\int_{-a}^a \text{odd } dt = 0$; $\int_{-a}^a \text{even} = 2 \int_0^a$.
Periodicity	$x(t) = x(t + T)$ for all t ; $x[n] = x[n + N]$	Smallest $T > 0$ (N integer) is the fundamental period; $\omega_0 = 2\pi/T$.
DT sinusoid period	$\cos(\omega_0 n)$ periodic iff $\frac{\omega_0}{2\pi} = \frac{m}{N}$ (rational)	Then period N (smallest). Sum of periodic signals: period = LCM.
Energy	$E = \int_{-\infty}^{\infty} x(t) ^2 dt$, $E = \sum_n x[n] ^2$	Total energy. Energy signal: $0 < E < \infty$ (then $P = 0$).
Power	$P = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x ^2 dt$; periodic: $P = \frac{1}{T} \int_T x ^2 dt$	Average power. Power signal: $0 < P < \infty$ (then $E = \infty$).
Energy under scaling	$E\{x(at + b)\} = E\{x\}/ a $	Shifts/flips keep energy; compressing by $ a $ divides it.

3 System properties

Name	Formula	Definition / meaning
Memoryless	$y(t)$ depends only on $x(t)$	LTI: $h[n] = c\delta[n]$.
Causal	$y(t_0)$ depends only on $x(t)$, $t \leq t_0$	LTI: $h[n] = 0$ for $n < 0$ ($h(t) = 0$, $t < 0$).
Linear	$ax_1 + bx_2 \rightarrow ay_1 + by_2$	Additivity + homogeneity (superposition). Zero in $\Rightarrow$ zero out.
Time-invariant	$x(t - t_0) \rightarrow y(t - t_0)$	Shifting the input only shifts the output.
BIBO stable	bounded $x \Rightarrow$ bounded y	LTI: $\sum_n h[n] < \infty$, $\int h(t) dt < \infty$.
Invertible	$\exists g: h * g = \delta$	Distinct inputs give distinct outputs.

4 Convolution and LTI systems

Name	Formula	Definition / meaning
Impulse response	$h[n] = T\{\delta[n]\}$	Output of the system when the input is an impulse; fully characterises an LTI system.
DT convolution	$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$	Output of an LTI system. Flip h , slide by n , multiply, add.
CT convolution	$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$	Continuous version (integral replaces sum).
Output length	$L + M - 1$; start index $n_x + n_h$	L, M = lengths of x, h ; n_x, n_h = their first indices.
Commutative	$x * h = h * x$	Input and impulse response can be swapped.
Associative (series)	$(x * h_1) * h_2 = x * (h_1 * h_2)$	Cascade $\Rightarrow h = h_1 * h_2$ (order doesn't matter).
Distributive (parallel)	$x * (h_1 + h_2) = x * h_1 + x * h_2$	Parallel branches added $\Rightarrow h = h_1 + h_2$.
Identity / shift	$x * \delta[n] = x[n]$, $x * \delta[n - n_0] = x[n - n_0]$	Convolving with a shifted impulse shifts the signal.
Step response	$s[n] = \sum_{k=-\infty}^n h[k]$, $s(t) = \int_{-\infty}^t h(\tau) d\tau$	Output for $x = u$; running sum of h . $h[n] = s[n] - s[n - 1]$.
Output for $u[n]$	$y[n] = \sum_k h[k] u[n - k]$	Weighted sum of shifted steps, one per tap of h .
Sum rule	$\sum_n y[n] = (\sum x) (\sum h)$	Quick check for convolution answers.
Cross-correlation	$r_{xy}[l] = \sum_n x[n] y[n - l] = x[n] * y[-n]$	Similarity at lag l ; peak location = delay. Out-of-phase signals: most negative peak.
Autocorrelation	$r_{xx}[l] = \sum_n x[n] x[n - l]$, $r_{xx}[0] = E$	Maximum at $l = 0$, even in l .
Eigenfunction	$e^{j\omega n} \rightarrow H(e^{j\omega})e^{j\omega n}$	Complex exponentials pass through LTI systems scaled by the frequency response.

5 Continuous-time Fourier series

Name	Formula	Definition / meaning
Synthesis	$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$	Periodic x as a sum of harmonics $k\omega_0$; $\omega_0 = 2\pi/T$.
Analysis	$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt$	a_k = complex amplitude of harmonic k (integrate over any one period).
DC coefficient	$a_0 = \frac{1}{T} \int_T x(t) dt$	Average value of the signal.
Trigonometric form	$x = A_0 + \sum_{n \geq 1} [A_n \cos n\omega_0 t + B_n \sin n\omega_0 t]$	$A_0 = a_0$, $A_n = \frac{2}{T} \int_T x \cos n\omega_0 t dt$, $B_n = \frac{2}{T} \int_T x \sin n\omega_0 t dt$.
Exp. $\leftrightarrow$ trig	$A_n = 2 \operatorname{Re} a_n$, $B_n = -2 \operatorname{Im} a_n$; $a_n = \frac{A_n - jB_n}{2}$	Conversion for real signals.
FT shortcut	$a_k = \frac{1}{T} X_1(jk\omega_0)$	X_1 = Fourier transform of one period only; sample it at the harmonics.
Square wave	$x = 1$ for $ t < T_1$: $a_k = \frac{\sin(k\omega_0 T_1)}{k\pi}$, $a_0 = \frac{2T_1}{T}$	Standard pair; T_1 is half the pulse width.
Impulse train	$\sum_n \delta(t - nT)$: $a_k = \frac{1}{T}$ for all k	All harmonics equal – the basis of sampling theory.
Linearity	$Ax + By \rightarrow Aa_k + Bb_k$	Same period T .
Time shift	$x(t - t_0) \rightarrow a_k e^{-jk\omega_0 t_0}$	Magnitudes unchanged, phases rotate.
Time reversal	$x(-t) \rightarrow a_{-k}$	
Time scaling	$x(\alpha t) \rightarrow a_k$ (period T/α)	Coefficients unchanged; fundamental becomes $\alpha\omega_0$.
Conjugation	$x^*(t) \rightarrow a_{-k}^*$	
Real signal	$a_{-k} = a_k^*$	$ a_k $ even, $\angle a_k$ odd.
Real & even	a_k real and even $\Rightarrow B_n = 0$	Only cosine terms.
Real & odd	a_k imaginary and odd $\Rightarrow A_0 = A_n = 0$	Only sine terms.
Half-wave symmetry	$x(t + T/2) = -x(t) \Rightarrow a_k = 0$ for even k	Only odd harmonics.
Differentiation	$\frac{dx}{dt} \rightarrow jk\omega_0 a_k$	Differentiate until impulses appear, then divide back.
Integration	$\int x dt \rightarrow \frac{a_k}{jk\omega_0}$ (needs $a_0 = 0$)	
Frequency shift	$e^{jM\omega_0 t} x(t) \rightarrow a_{k-M}$	
Periodic convolution	$\int_T x(\tau) y(t - \tau) d\tau \rightarrow T a_k b_k$	
Multiplication	$x(t)y(t) \rightarrow \sum_l a_l b_{k-l}$	Discrete convolution of coefficients.
Parseval	$\frac{1}{T} \int_T x(t) ^2 dt = \sum_k a_k ^2$	Average power = sum of powers of harmonics.
Dirichlet / Gibbs	Converges to midpoint at jumps; overshoot $\approx 9\%$	Finite partial sums ring near discontinuities.

6 Continuous-time Fourier transform (CTFT)

Name	Formula	Definition / meaning
Analysis	$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$	Spectrum of an aperiodic signal; ω in rad/s. (Hz form: $X(f)$, $\omega = 2\pi f$.)
Synthesis	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)e^{j\omega t} d\omega$	Inverse transform.
DC value	$X(0) = \int x(t)dt$; $x(0) = \frac{1}{2\pi} \int X d\omega$	Area checks.
Magnitude / phase	$X = X e^{j\angle X}$	$ X $ amplitude spectrum, $\angle X$ phase spectrum.
Linearity	$ax + by \rightarrow aX + bY$	
Time shift	$x(t - t_0) \rightarrow e^{-j\omega t_0} X(j\omega)$	Adds linear phase $-\omega t_0$.
Frequency shift	$e^{j\omega_0 t} x(t) \rightarrow X(j(\omega - \omega_0))$	Modulation moves the spectrum.
Modulation by cosine	$x(t) \cos \omega_0 t \rightarrow \frac{1}{2}[X(j(\omega - \omega_0)) + X(j(\omega + \omega_0))]$	Two half-height copies at $\pm\omega_0$.
Time scaling	$x(at) \rightarrow \frac{1}{ a } X\left(\frac{j\omega}{a}\right)$	Compress in time $\Leftrightarrow$ stretch in frequency.
Combined	$x(at + b) \rightarrow \frac{1}{ a } e^{j\omega b/a} X\left(\frac{j\omega}{a}\right)$	Apply scaling then the shift $-b/a$.
Time reversal	$x(-t) \rightarrow X(-j\omega)$	
Conjugation	$x^*(t) \rightarrow X^*(-j\omega)$	For real x : $X(-j\omega) = X^*(j\omega)$.
Duality	$X(jt) \rightarrow 2\pi x(-\omega)$	Time and frequency pairs swap roles.
Differentiation in time	$\frac{dx}{dt} \rightarrow j\omega X$	
Integration	$\int_{-\infty}^t x d\tau \rightarrow \frac{X}{j\omega} + \pi X(0)\delta(\omega)$	
Differentiation in freq.	$t x(t) \rightarrow j \frac{dX}{d\omega}$	
Convolution	$x * h \rightarrow X(j\omega)H(j\omega)$	LTI filtering = multiplying spectra; H is the frequency response.
Multiplication	$x(t)p(t) \rightarrow \frac{1}{2\pi} X * P$	Basis of sampling and modulation.
Parseval	$\int x(t) ^2 dt = \frac{1}{2\pi} \int X(j\omega) ^2 d\omega$	Energy is the same in both domains; $ X ^2$ = energy spectral density.
Symmetry	real & even $x \Rightarrow X$ real & even; real & odd $\Rightarrow X$ imaginary & odd	

Standard pairs ($a > 0$)

$\delta(t) \leftrightarrow 1$, $1 \leftrightarrow 2\pi\delta(\omega)$	Impulse has a flat spectrum; DC is an impulse at 0.
$\delta(t - t_0) \leftrightarrow e^{-j\omega t_0}$	
$e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0)$	Single spectral line.
$\cos \omega_0 t \leftrightarrow \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$	
$\sin \omega_0 t \leftrightarrow \frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$	
$e^{-at}u(t) \leftrightarrow \frac{1}{a + j\omega}$	One-sided exponential. $te^{-at}u(t) \leftrightarrow \frac{1}{(a + j\omega)^2}$.
$e^{-a t } \leftrightarrow \frac{2a}{a^2 + \omega^2}$	Two-sided exponential.
$e^{-at^2} \leftrightarrow \sqrt{\pi/a} e^{-\omega^2/4a}$	Gaussian $\rightarrow$ Gaussian. $e^{-t^2} \leftrightarrow \sqrt{\pi}e^{-\omega^2/4}$.

Name	Formula	Definition / meaning
	$x = 1, t \leq T_1 \leftrightarrow \frac{2 \sin \omega T_1}{\omega}$	Rect $\rightarrow$ sinc.
	$\frac{\sin Wt}{\pi t} \leftrightarrow 1, \omega < W$	Sinc $\rightarrow$ rect (ideal low-pass).
	$1 - t /\tau, t \leq \tau \leftrightarrow \tau \frac{\sin^2(\omega\tau/2)}{(\omega\tau/2)^2}$	Triangle $\rightarrow$ sinc ² .
	$u(t) \leftrightarrow \frac{1}{j\omega} + \pi\delta(\omega)$	
	$\sum_n \delta(t - nT) \leftrightarrow \frac{2\pi}{T} \sum_k \delta(\omega - \frac{2\pi k}{T})$	Impulse train $\rightarrow$ impulse train.
Periodic signal	$x = \sum a_k e^{jk\omega_0 t} \leftrightarrow \sum 2\pi a_k \delta(\omega - k\omega_0)$	Spectrum of a periodic signal is a set of lines.
Laplace link	$X(s) _{s=\sigma+j\omega} = \mathcal{F}\{x(t)e^{-\sigma t}\}$	Laplace transform = CTFT of the exponentially weighted signal.

7 Discrete-time Fourier transform (DTFT)

Name	Formula	Definition / meaning
Analysis	$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$	Spectrum of a DT signal; ω in rad/sample.
Synthesis	$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$	
Periodicity	$X(e^{j(\omega+2\pi)}) = X(e^{j\omega})$	Only $-\pi < \omega \leq \pi$ is unique; $\omega = \pi$ is the highest frequency.
Link to analog freq.	$\omega = \Omega T = 2\pi f / f_s$	Ω analog (rad/s), T sampling period.
Shift	$x[n - n_0] \rightarrow e^{-j\omega n_0} X$	
Modulation	$e^{j\omega_0 n} x[n] \rightarrow X(e^{j(\omega-\omega_0)});$ $(-1)^n x[n] \rightarrow X(e^{j(\omega-\pi)})$	$(-1)^n$ swaps low and high frequencies.
Convolution	$x * h \rightarrow X \cdot H$	
Parseval	$\sum x[n] ^2 = \frac{1}{2\pi} \int_{2\pi} X ^2 d\omega$	
Pair	$a^n u[n] \leftrightarrow \frac{1}{1 - ae^{-j\omega}} (a < 1)$	
z-transform link	$X(z) _{z=re^{j\omega}} = \text{DTFT}\{x[n]r^{-n}\}$	z-transform = DTFT of the weighted sequence.

8 Discrete Fourier transform (DFT)

Name	Formula	Definition / meaning
Twiddle factor	$W_N = e^{-j2\pi/N}$	N -th root of unity; $W_N^N = 1$, $W_N^{N/2} = -1$, $W_N^{N/4} = -j$.
DFT	$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, k = 0..N-1$	N samples in $\rightarrow N$ frequency bins out.
IDFT	$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}$	Inverse; note the $1/N$.
DFT = sampled DTFT	$X[k] = X(e^{j\omega}) _{\omega=2\pi k/N}$	For a length- $\leq N$ sequence.

Name	Formula	Definition / meaning
DFT and DTFS	$X[k] = N a_k$	a_k = DT Fourier series coefficients of the periodic extension.
DFT and CT series	$c_k \approx \frac{1}{N} X[k]$ (samples of one period)	Approximating CT Fourier coefficients from N samples of one period.
Bin frequency	$f_k = k \frac{f_s}{N}$	Analog frequency of bin k ($k \leq N/2$); bins $N/2 < k < N$ are negative freqs $(k - N)f_s/N$.
Resolution	$\Delta f = \frac{f_s}{N} = \frac{1}{NT}$	Set by record duration NT ; zero-padding doesn't improve it.
DC and Nyquist bins	$X[0] = \sum x[n]; X[N/2] = \sum (-1)^n x[n]$	
4-point DFT	$X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} x$	Row k uses $(-j)^{kn}$. Example $\{1, 2, 3, 4\} \rightarrow \{10, -2 + 2j, -2, -2 - 2j\}$.
Periodicity	$X[k + N] = X[k], x[n + N] = x[n]$ (implied)	DFT treats the data as one period.
Modulo index	$((n))_N = n \bmod N$	e.g. $((-1))_8 = 7, ((9))_8 = 1$.
Linearity	$ax + by \rightarrow aX + bY$	Same length N (pad first).
Circular shift	$x[((n - m))_N] \rightarrow W_N^{km} X[k]$	Rotation; magnitude unchanged.
Modulation	$W_N^{-k_0 n} x[n] \rightarrow X[((k - k_0))_N]$	$(-1)^n x[n] \rightarrow X[((k - N/2))_N]$.
Time reversal	$x[((-n))_N] \rightarrow X[((-k))_N]$	
Conjugation	$x^*[n] \rightarrow X^*[((-k))_N]$	
Conjugate symmetry	real $x \Rightarrow X[N - k] = X^*[k]$	$ X $ even, $\angle X$ odd about $N/2$; only bins $0..N/2$ are free.
Duality	$X[n] \rightarrow N x[((-k))_N]$	
Circular convolution	$(x \circledast h)[n] = \sum_{m=0}^{N-1} x[m] h[((n - m))_N]$	Convolution on a clock of N points.
Convolution theorem	$x \circledast h \leftrightarrow X[k] H[k]$	Multiplying DFTs gives <i>circular</i> , not linear, convolution.
Multiplication	$x[n] h[n] \leftrightarrow \frac{1}{N} X \circledast H$	
Circular vs linear	$y_c[n] = \sum_r y_{\text{lin}}[n + rN]$	Circular result = linear result folded (time-aliased) mod N .
No-wrap condition	$N \geq L + M - 1$	Zero-pad both to N ; then circular = linear.
Parseval (DFT)	$\sum_n x[n] ^2 = \frac{1}{N} \sum_k X[k] ^2$	$ X[k] ^2/N$ = energy in bin k .
Zero-padding	append zeros: $N \rightarrow N'$	Samples the same DTFT more densely (smoother plot), no new resolution; changes the period of a periodic signal.
Leakage	tone not an integer number of cycles in N	Energy spreads to neighbouring bins; reduce with windows or a better N .

9 Fast Fourier transform (FFT)

Name	Formula	Definition / meaning
Direct DFT cost	N^2 mults, $N(N-1)$ adds	$O(N^2)$.
Radix-2 FFT cost	$\frac{N}{2} \log_2 N$ mults, $N \log_2 N$ adds	$O(N \log N)$; needs $N = 2^m$. $\log_2 N$ stages, $N/2$ butterflies each.
DIT split	$G[k] = \sum_r x[2r] W_{N/2}^{rk}, H[k] = \sum_r x[2r+1] W_{N/2}^{rk}$	$G = N/2$ -point DFT of even samples, $H =$ of odd samples.
DIT butterfly	$X[k] = G[k] + W_N^k H[k],$ $X[k + \frac{N}{2}] = G[k] - W_N^k H[k]$	$0 \leq k < N/2$. Inputs in bit-reversed order, outputs in natural order.
Key identities	$W_N^2 = W_{N/2}, W_N^{k+N/2} = -W_N^k$	Why the split works.
DIF split	$a[n] = x[n] + x[n + \frac{N}{2}],$ $b[n] = (x[n] - x[n + \frac{N}{2}]) W_N^n$	$X[2m] = \text{DFT}_{N/2}\{a\},$ $X[2m+1] = \text{DFT}_{N/2}\{b\}$. Inputs natural, outputs bit-reversed.
2-point butterfly	$X[0] = x_0 + x_1, X[1] = x_0 - x_1$	Smallest DFT.
Bit reversal	index $b_{m-1} \dots b_1 b_0 \rightarrow b_0 b_1 \dots b_{m-1}$	$N = 8$ order: 0, 4, 2, 6, 1, 5, 3, 7.
Cooley–Tukey ($N = N_1 N_2$)	$X[N_2 k_1 + k_2] = \sum_{n_1} W_{N_1}^{n_1 k_1} W_N^{n_1 k_2} \sum_{n_2} x[N_1 n_2 + n_1] W_{N_2}^{n_2 k_2}$	Input column-major $n = N_1 n_2 + n_1$, output row-major $k = N_2 k_1 + k_2$.
Bailey 4-step	rows $\rightarrow$ twiddle $W_N^{n_1 k_2} \rightarrow$ columns $\rightarrow$ read	(1) $N_1 \times N_2$ matrix; (2) N_2 -pt DFT of rows; (3) twiddle; (4) N_1 -pt DFT of columns.
Radix-3	$X[k + mN/3] = \sum_{r=0}^2 W_3^{rm} W_N^{rk} G_r[k]$	$G_r =$ DFT of samples $n \equiv r \pmod{3}$.
Bluestein identity	$kn = \frac{1}{2}[k^2 + n^2 - (k-n)^2]$	Turns the DFT into a convolution (any N , including prime).
Bluestein DFT	$X[k] = W_{2N}^{k^2} (a * b)[k], a[n] = x[n] W_{2N}^{n^2},$ $b[n] = W_{2N}^{-n^2}$	Chirp convolution computed with FFTs of size $M \geq 2N - 1$ ($M = 2^m$); negative b indices wrap to the end.
Fast convolution	$y = \text{IFFT}\{\text{FFT}(x, N) \cdot \text{FFT}(h, N)\},$ $N \geq L + M - 1$	Cost $\approx 3N \log_2 N$ vs LM direct.

10 Sampling and reconstruction

Name	Formula	Definition / meaning
Sampling period / rate	$T = \frac{1}{f_s}, \omega_s = \frac{2\pi}{T} = 2\pi f_s$	T seconds between samples; f_s samples/s.
Samples	$x[n] = x(nT)$	Values of the CT signal at the sampling instants.
Impulse train	$p(t) = \sum_n \delta(t - nT)$	Sampling function.
Sampled signal	$x_p(t) = x(t)p(t) = \sum_n x(nT)\delta(t - nT)$	Impulses weighted by the samples.
Spectrum of p	$P(j\omega) = \frac{2\pi}{T} \sum_k \delta(\omega - k\omega_s)$	From $a_k = 1/T$ for the impulse train.
Sampled spectrum	$X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s))$	Copies of X every ω_s , scaled by $1/T$.

Name	Formula	Definition / meaning
DTFT of samples	$X(e^{j\omega}) = X_p(j\omega/T)$	Same spectrum with $\omega = \Omega T$.
Band-limited	$X(j\omega) = 0$ for $ \omega > \omega_M$	ω_M (or f_M) = highest frequency present.
Nyquist theorem	$\omega_s > 2\omega_M$ ($f_s > 2f_M$)	Then $x(t)$ is uniquely recovered from its samples.
Nyquist rate	$2\omega_M$ (or $2f_M$)	Minimum sampling rate required (property of the signal).
Nyquist / folding frequency	$\omega_s/2$ (or $f_s/2$)	Highest frequency representable at a given f_s (property of the sampler).
Guard band	$\omega_s - 2\omega_M$	Gap between spectral copies; eases the reconstruction filter.
Alias frequency	$f_a = f_0 - kf_s $, $k = \text{round}(f_0/f_s)$	Apparent frequency of a tone $f_0 > f_s/2$; lies in $[0, f_s/2]$.
Alias sign rule	$f_0 - kf_s < 0$: $\cos(\cdot + \phi) \rightarrow \cos(2\pi f_a t - \phi)$, $\sin \rightarrow -\sin$	Folding from the negative side flips the phase / the sign of a sine.
Bandwidth of products	$\cos \omega_1 t \cos \omega_2 t \rightarrow \omega_1 \pm \omega_2$; $x^2(t)$: $2\omega_M$	Multiplication widens bandwidth; convolution keeps the smaller one.
Reconstruction	$x_r(t) = \sum_n x(nT) h(t - nT)$	Filter the impulse train with $h(t)$; $X_r = X_p \cdot H$.
Ideal LPF	$H = T$ for $ \omega < \omega_c$, $\omega_M < \omega_c < \omega_s - \omega_M$	Usually $\omega_c = \omega_s/2$. Gain T cancels the $1/T$.
Ideal interpolation	$x_r(t) = \sum_n x(nT) \frac{\omega_c T}{\pi} \frac{\sin \omega_c(t - nT)}{\omega_c(t - nT)}$	With $\omega_c = \omega_s/2$: $x_r = \sum x(nT) \text{sinc}(\frac{t - nT}{T})$. Non-causal, infinite.
Zero-order hold	$h_0(t) = 1$, $0 \leq t < T$; $H_0 = T e^{-j\omega T/2} \frac{\sin(\omega T/2)}{\omega T/2}$	Staircase output. Gain at $\omega_s/2$: $2T/\pi$ (0.637 of DC). Sidelobes decay $\propto 1/\omega$.
ZOH relative gain	$ H_0(f) /T = \left \frac{\sin(\pi f/f_s)}{\pi f/f_s} \right $	Attenuation (droop) of a component at f .
Linear interp. (FOH)	$h_1(t) = 1 - \frac{ t }{T}$, $ t \leq T$; $H_1 = T \frac{\sin^2(\omega T/2)}{(\omega T/2)^2}$	Joins samples with straight lines; $h_1 = \frac{1}{T} h_0 * h_0$; decays $\propto 1/\omega^2$.
Linear interp. value	$x(t) \approx x_a + \frac{t - t_a}{t_b - t_a} (x_b - x_a)$	Value between neighbouring samples (t_a, x_a) and (t_b, x_b) .
Downsampling by M	$y[n] = x[Mn]$, new rate f_s/M	Needs anti-alias LPF (cutoff π/M) first.
Upsampling by L	insert $L - 1$ zeros, then interpolate; rate Lf_s	Interpolation filter removes images.
Rational rate change	rate $\times \frac{L}{M}$: up by L , filter, down by M	e.g. $3 \rightarrow 2$ samples/s: $L = 2$, $M = 3$.
Anti-aliasing filter	LPF with cutoff $\leq f_s/2$ before sampling	Real (time-limited) signals are never exactly band-limited.
Time-frequency duality	sample in time $\leftrightarrow$ periodic in frequency	Sampling in frequency (DFT) $\leftrightarrow$ periodic in time.

11 NumPy and Matplotlib reference

Name	Formula	Definition / meaning
Array creation	<code>np.array</code> , <code>np.arange(a,b,s)</code> , <code>np.linspace(a,b,n)</code> , <code>np.zeros(n)</code>	<code>arange</code> excludes b ; <code>linspace</code> includes b .
Slicing	<code>a[start:stop:step]</code> , <code>a[::-1]</code> , <code>a[:,1]</code>	Stop is exclusive; negative step reverses.
Boolean mask	<code>a[a>5]</code> , <code>a[(a>2)&(a<8)]</code>	Returns a 1-D array of elements where the condition is true.
Where	<code>np.where(cond, A, B)</code>	Elementwise choice.
Reductions	<code>sum</code> , <code>mean</code> , <code>max</code> , <code>argmax</code> , <code>cumsum</code> , <code>count_nonzero</code>	<code>axis=0</code> down columns, <code>axis=1</code> across rows.
Reshape / transpose	<code>a.reshape(r,c)</code> , <code>a.T</code>	Fills row by row.
Convolution	<code>np.convolve(x,h)</code>	Linear convolution, length $L + M - 1$.
Correlation	<code>np.correlate(x,y,'full')</code>	Lag of peak = index $-(\text{len}(y) - 1)$.
FFT	<code>np.fft.fft(x, N)</code> , <code>np.fft.ifft(X)</code>	Pads/truncates to N ; output complex.
Frequency axis	<code>np.fft.fftfreq(N, d=1/fs)</code> , <code>np.fft.fftshift</code>	Bin frequencies; shift puts 0 Hz in the middle.
Magnitude / phase	<code>np.abs(X)</code> , <code>np.angle(X)</code>	
Plots	<code>plt.plot (CT)</code> , <code>plt.stem (DT)</code> , <code>plt.subplot</code>	Label axes, add grid.
Print format	ints [1 2 3], floats [1. 2.5], mean 5.0	/ always gives floats; // floor division.

12 Useful mathematics

Name	Formula	Definition / meaning
Geometric sum	$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a}$; $\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$ ($ a < 1$)	Used for DTFT/DFT of exponentials.
Orthogonality	$\sum_{n=0}^{N-1} W_N^{kn} = N$ if $k \equiv 0 \pmod{N}$, else 0	Why the IDFT inverts the DFT.
CT orthogonality	$\int_T e^{j(k-m)\omega_0 t} dt = T\delta_{km}$	Why the FS analysis formula works.
Gaussian integral	$\int_{-\infty}^{\infty} e^{-at^2} dt = \sqrt{\pi/a}$	
Lorentzian integral	$\int_{-\infty}^{\infty} \frac{d\omega}{a^2 + \omega^2} = \frac{\pi}{a}$	For Parseval with $e^{-at}u(t)$.
Integration by parts	$\int u dv = uv - \int v du$	$\int te^{-j\omega t} dt$ in Fourier series.
Exponential integral	$\int_a^b e^{-j\omega t} dt = \frac{e^{-j\omega a} - e^{-j\omega b}}{j\omega}$	Building block of every FT/FS calculation.
Trig identities	$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$; $1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$	Simplifying spectra.
Complex numbers	$ a + jb = \sqrt{a^2 + b^2}$, $\angle = \tan^{-1} \frac{b}{a}$, $(a + jb)^* = a - jb$	$j^2 = -1$, $\frac{1}{j} = -j$, $(-j)^2 = -1$.

Name	Formula	Definition / meaning
Basel sums	$\sum_{k \geq 1} \frac{1}{k^2} = \frac{\pi^2}{6}, \sum_{k \text{ odd}} \frac{1}{k^2} = \frac{\pi^2}{8}$	Appear from Parseval on sawtooth/square waves.